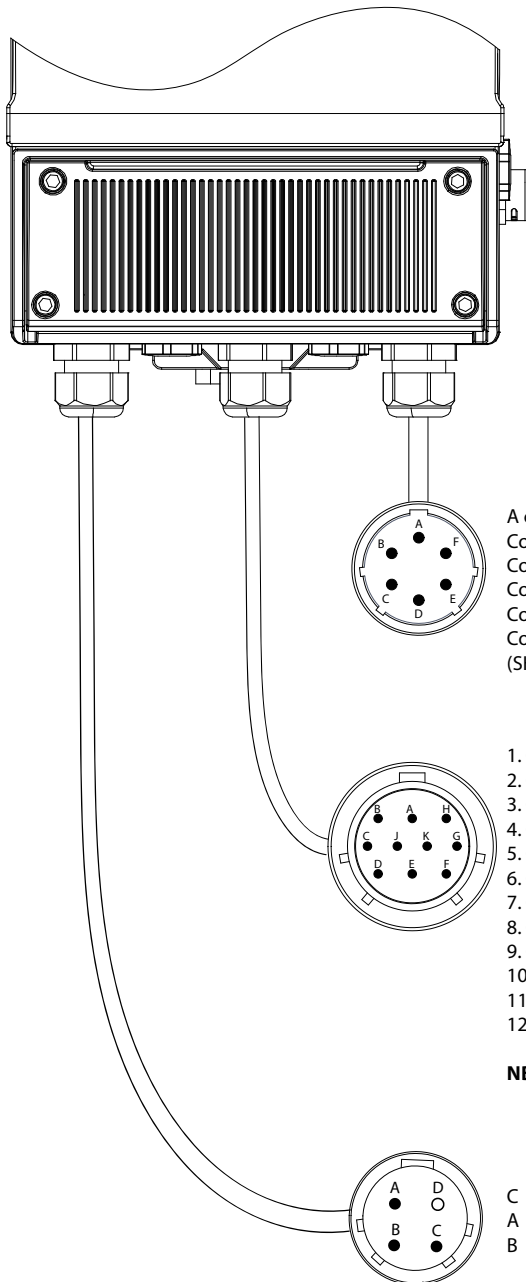


CONNECTORS MIL

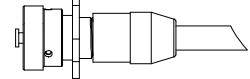
THE following are the links of the MIL connectors IP68



SENSOR SIGNALS

- A contact from terminal 1 of the converter (electrode 1)
- Contact F from terminal 2 of the converter (electrode 2)
- Contact And from terminal 3 of the converter (COM. Elec.)
- Contact B from terminal 13 of the converter (COIL 1)
- Contact C from terminal 12 of the converter (COIL 2)
- Contact D from terminal 4 and 11 of converter (SHIELD electrodes - COILS)

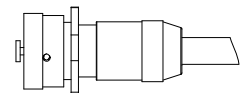
Connector
6 poles



INPUT/OUTPUT

1. Contact B from terminal 24 of the converter (Out1 4-20 +)
2. Contact A from terminal 25 of the converter (Out2 4-20 +)
3. Contact H from terminal 23 of the converter (Out1-2 4-20 -)
4. Contact C from terminal 21 of the converter (+ 24V)
5. Contact J from terminal 22 of the converter (0V)
6. Contact K from terminal 15 of the inverter (IN1 +)
7. Contact G from terminal 16 of the converter (IN1-)
8. Contact D from terminal 19 of the converter (C Out2)
9. Contact E from terminal 18 of the converter (Out1 C)
10. Contact F from terminal 17 of the converter (Out1-2 E)
11. Contact C from terminal 28 of the converter (RS485 A)
12. Contact J from terminal 29 of the converter (RS485 B)

Connector
10 poles

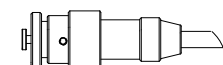


NB: the connections 4 and 5 exclude the use connections 11 and 12.

POWER SUPPLY

- C from terminal L power converter (+ dc)
- A from terminal N power converter (-dc)
- B from terminal GROUND Power Converter

Connector
4 poles



NOTE: Military Connector 6 poles for sensor converter only provided in the separate version of the converter.

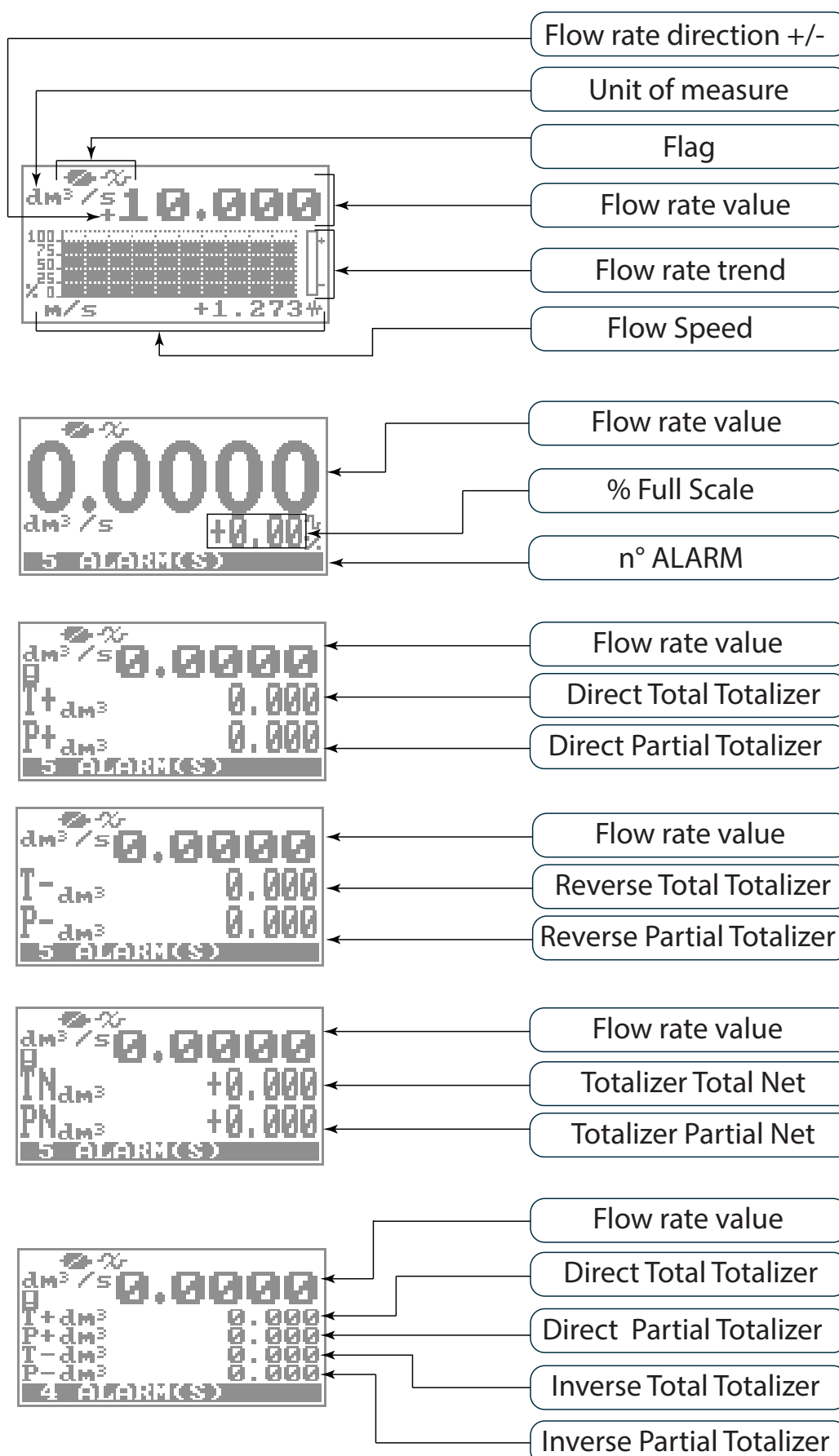
START VISUALIZATION PAGES



The direct exposure of the converter to the solar rays, could damage the liquid crystal display. The visualization pages can be change according to instrument's setup.

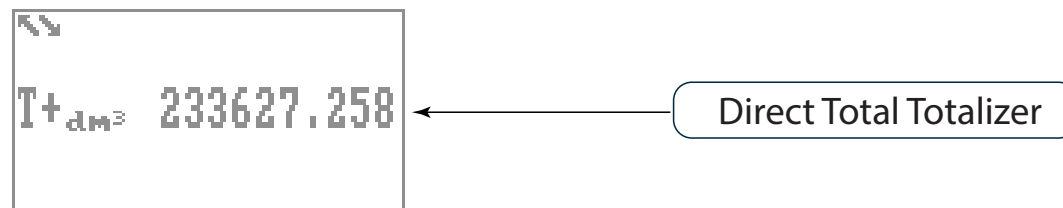
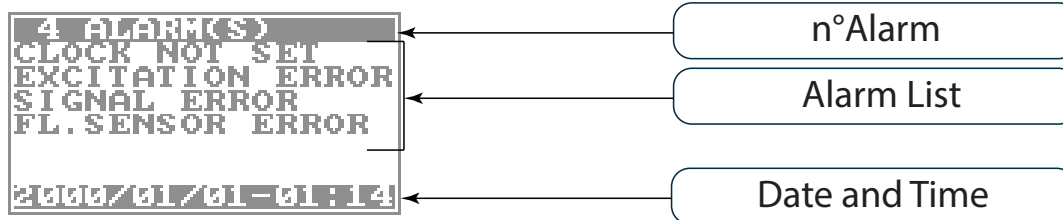
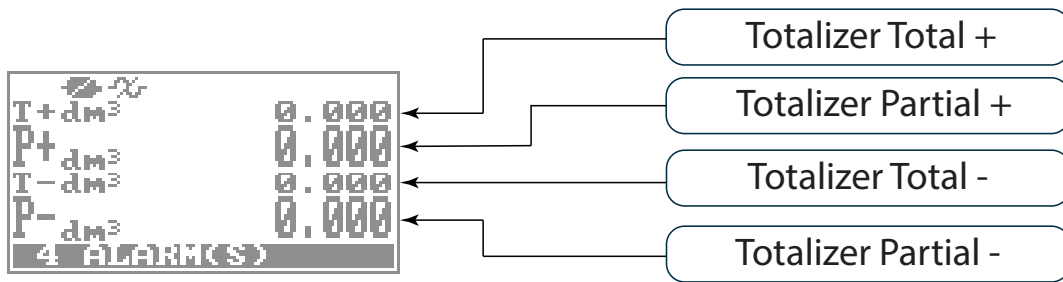


Push to change visualization





Push to change visualization

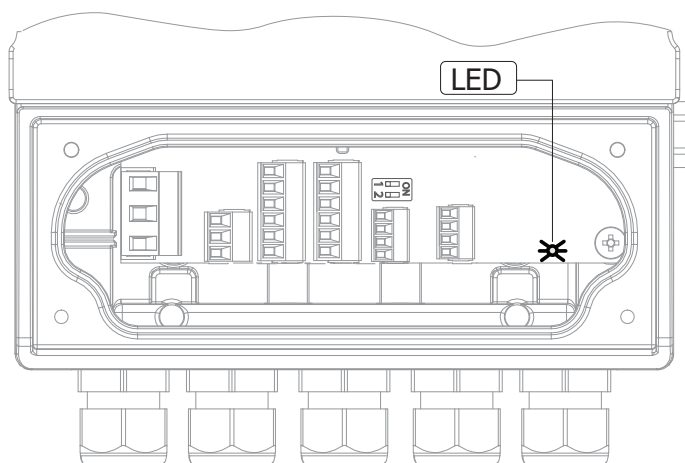


[The visualisation page about batching is described in the dedicated section "BATCHING" pag45](#)

MEANING OF FLAGS

FLAG	DESCRIPTION	FLAG	DESCRIPTION
	EMPTY PIPE		MIN FLOW ALARM
	FILE UPLOAD		MAX FLOW ALARM
	FILE DOWNLOAD		VIDEO TERMINAL CONNECTED
	BATTERY RECHARGE (FLASHING) LOW BATTERY (FIXED)		FLOW RATE OVERFLOW
	FLOW RATE SIMULATION (FLASHING)		PULSE 1 OVERFLOW
	CALIBRATION (FLASHING)		PULSE 2 OVERFLOW
	GENERIC ALARM (FLASHING)	POWERED DEVICE WITH ONE CHARGERS BATTERY (MID-DIRECTIVE) oppure BATCHING IN PROGRESS	
	GENERAL ALARM ONLY ON PHYSICAL DISPLAY (FLASHING)		
	SIGNAL ERROR		
	EXCITATION ERROR		

MEANING OF LED COLORS



LED Red: Alarm signal

LED Blue: Usb communication enable

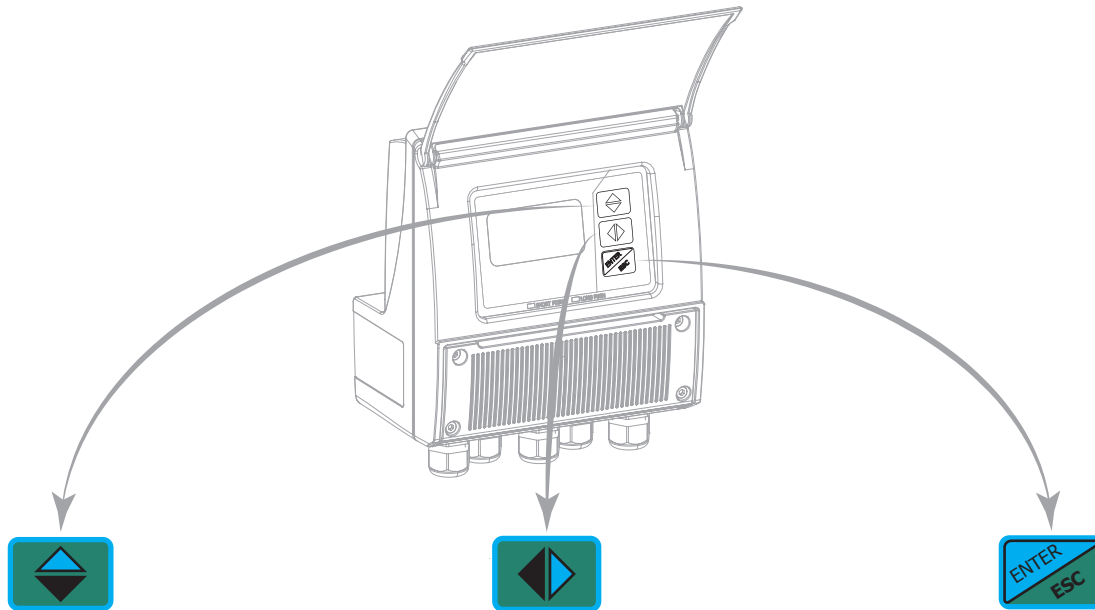
LED Green: Functioning system correctly

ACCESS TO THE CONFIGURATION MENU

The configuration can be done in two different ways:

- ☐ By keypad of converter
- ☐ By MCP interface (Virtual display of instrument)

ACCESS VIA KEYPAD



SHORT PRESSING (< 1 SECOND):

Increases the numeric figure or the parameter selected by the cursor
Returns to the previous subject on the menu.

LONG PRESSING (> 1 SECOND):

Decreases the numeric figure or the parameter selected by the cursor.
Proceeds to the next subject on the menu.

SHORT PRESSING (< 1 SECOND):

Moves/positions the cursor rightward on the input field. Proceeds to the following subject of the menu. Change the display of the process data

LONG PRESSING (> 1 SECOND):

Moves/positions the cursor leftward on the input field. Returns to the previous subject on the menu

SHORT PRESSING (< 1 SECOND):

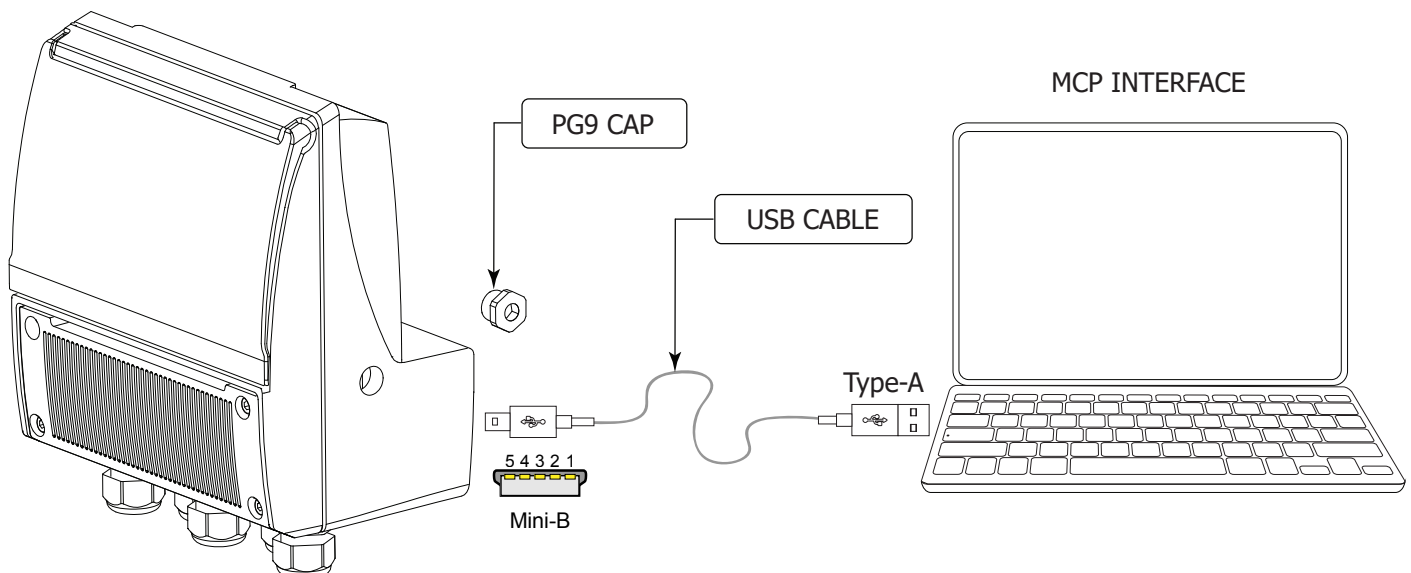
Enter /leave the selected function
Enables the main menu for the instrument configuration Cancels the selected function under progress

LONG PRESSING (> 1 SECOND):

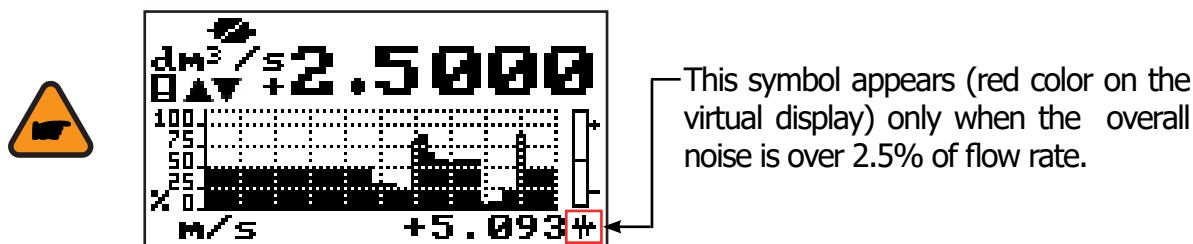
Leaves the current menu
Enables the totalizer reset request (when enabled) Confirms the selected function.

ACCESS VIA MCP INTERFACE (VIRTUAL DISPLAY)

MCP is a Windows® software that allows to set all the converter functions and personalize the menu. The MCP program is required for the blind version of the converter. To use MCP interface consult the relevant user manual.



FLOW RATE VISUALIZATION



The MV 210 can show a 5 digits display for flow rate units; this mean the maximum flow rate value that can be represented on the display is 99999 (no matter the positioning of the decimal point). The minimum is 0.0025. The representable measure unit depends on sensor flow rate and diameter; the permitted units are those, that permits the instrument full scale value not exceeding 99999.

Example for DN 300, Full scale value: 3m/s:

❑ **PERMITTED measure unit** (example): l/s (216.00); m³/h (777.60); m³/s (0.2160)

❑ **NOT PERMITTED measure unit** (example): l/h (777600)

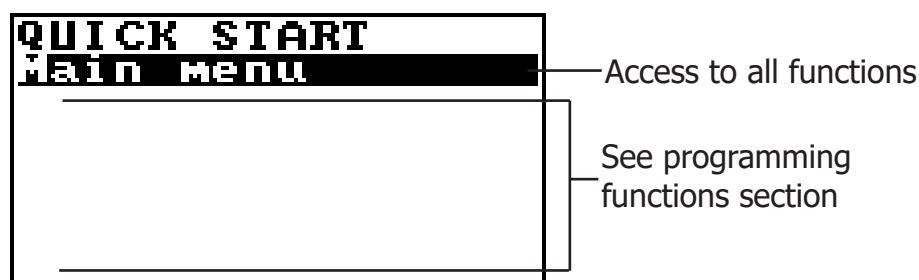
FLOW RATE ALERT



This FLAG becomes active when there is a flow variation (flow rate not stable).

QUICK START MENU

The QUICK START MENU allows to user immediate access to some of the most commonly used functions; through MCP software it possible customize this menu to make it suitable for the specific application.



The user has immediate access to the Quick Start menu when the converter is powered up by pressing the Enter key. If access to the quick start menu does not occur, then it could be disabled using the function "9.11" page 29 .

CONVERTER ACCESS CODE

The access for programming the instrument is regulated by six access levels logically grouped. Every level is protected by a different code.

❑ Access Level 1-2-3-4 Freely programmable by user

ACCESS CODE SET : MENU 13 SYSTEM

```

SYSTEM
Data.saving= ON
Time zone=h+01.00
2016/04/04-16:07
L1 code=*****
L2 code=*****
L3 code=*****
L4 code=*****
L5 code=*****
L6 code=*****
Restr.access= ON
010.011.012.013
010.011.012.014
255.255.255.000
KI= 0.96469
KS= 1.00000
KR= 1.00000
DAC1 4mA= 02460
DAC1 20mA= 11050
DAC2 4mA= 02460
DAC2 20mA= 11050
Stand-by
FW update
13-System
  
```

```

SYSTEM
L1 code=*****
L2 code=*****
L3 code=*****
L4 code=*****
L5 code=*****
L6 code=*****
0499999999
  
```

The CODE is Settable by keyboard or MCP interface.
Depending on the level of access different display functions will be visible. (See section "FUNCTIONS DESCRIPTION" page 32)
These access levels interact with the "Restricted access"

RESTRICTED ACCESS SET : MENU 13 SYSTEM

```

SYSTEM
L1 code=*****
L2 code=*****
L3 code=*****
L4 code=*****
L5 code=*****
L6 code=*****
Restr.access=OFF
  
```

Settable Values

ON

OFF

Restrict = ON: Access permitted only to functions provided for a specific level;

Example: If the operator has a code of access level 3, after having set it, he can change only the functions with level 3 access.

Restict = OFF: It enables to change functions for the selected level and ALL the functions with lower access level.

Example: If the operator has the code of level 3, after having set it, he can change all the functions at level 3 and those at lower levels.

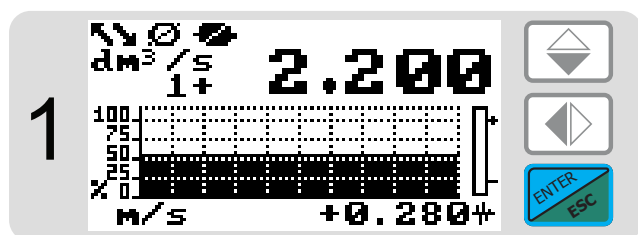
* WARNING: take careful note of the customized code, since there is no way for the user to retrieve or reset it if lost.

Factory preset access codes:

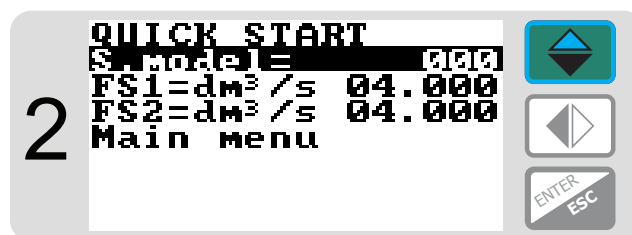
- ❑ L1: 10000000
- ❑ L2: 20000000
- ❑ L3: 30000000
- ❑ L4: 40000000

The following example shows how to change the Full scale by Quick Start menu; the second illustrates how to change the function by the Main menu.

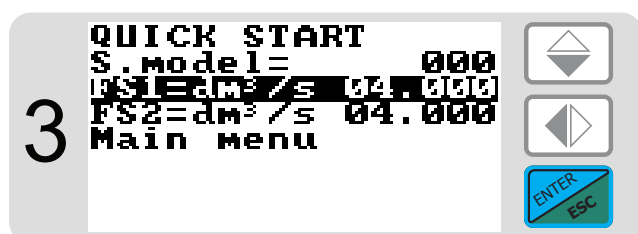
EXAMPLE: modifying the full scale value from 4dm³/s to 5dm³/s, from the "Quick start menu"



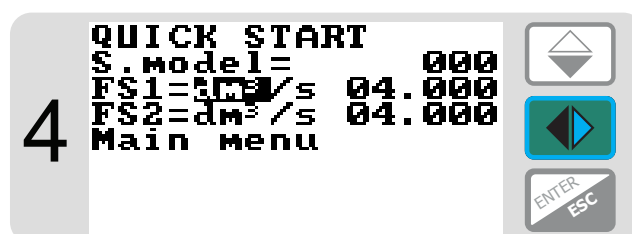
Press the ENTER button to access the Quick Start menu



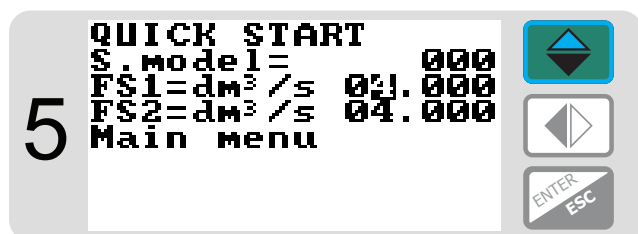
Select this function in the list to be edited



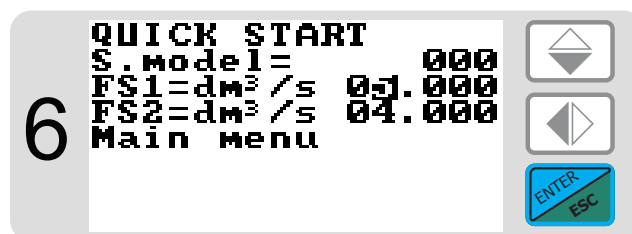
Press the ENTER button to select the function.



Select the value to be changed



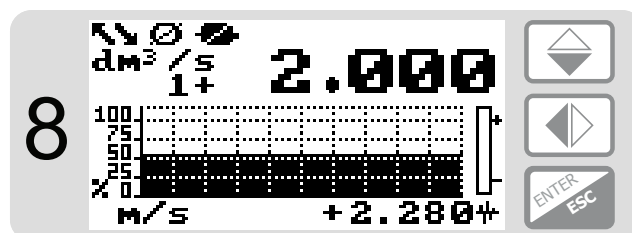
Change the value



Confirm the new value

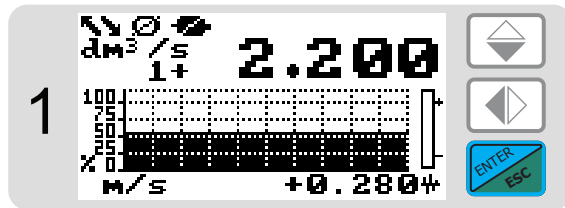


Long Push

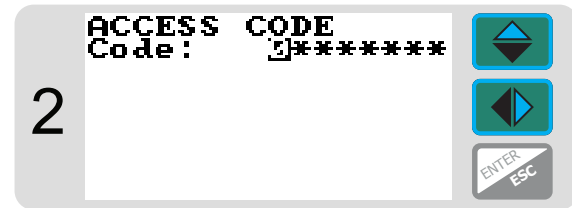


Main Page

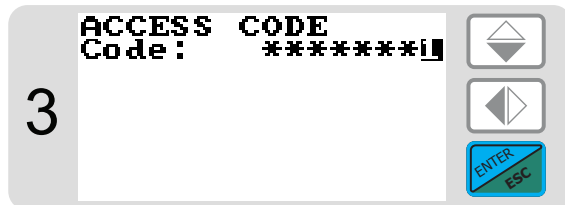
EXAMPLE: modifying the full scale value from 4dm³/s to 5dm³/s, from the "Main Menu" (quick start menu enabled)



Press the ENTER button to access the Quick Start menu



Press arrow keys to select the cell in which to insert the number of the access code.



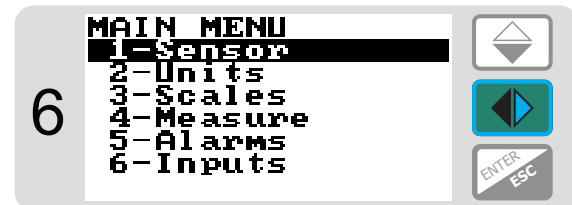
Press ENTER button to confirm value.



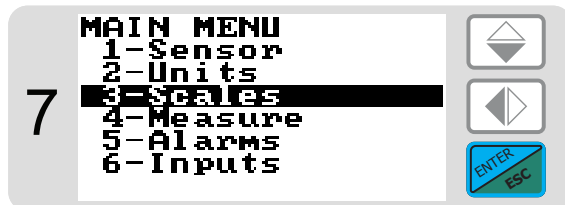
Select "Main Menu"



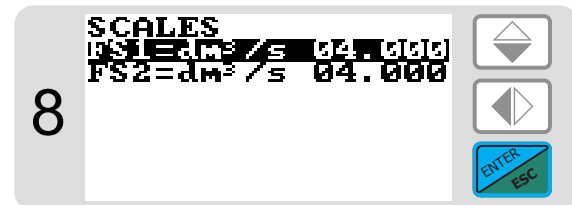
Press the ENTER button to access the Main Menu



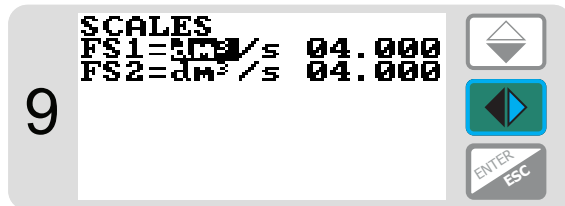
Select function



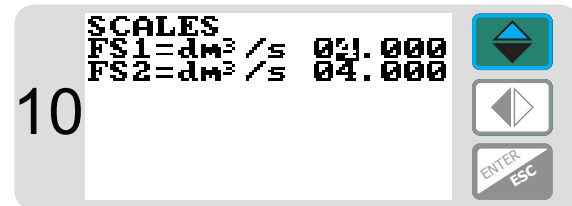
Press the ENTER button to access the "Scale Menu"



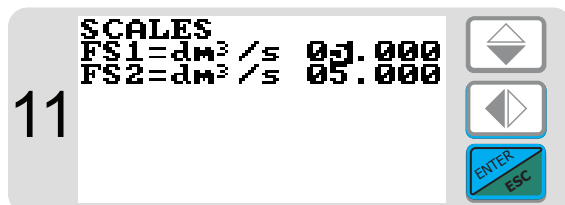
Press the ENTER button to access the "Fs1"



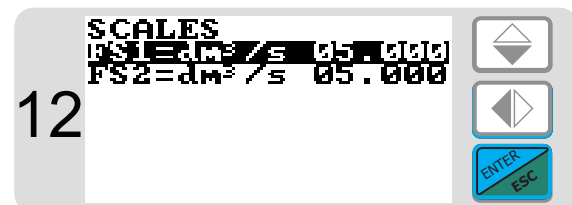
Select the value to be changed



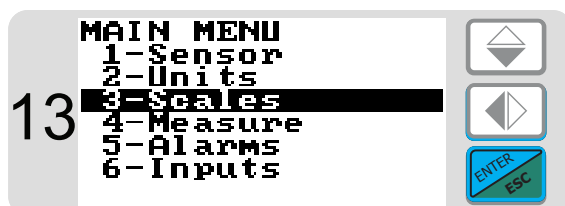
Change the value



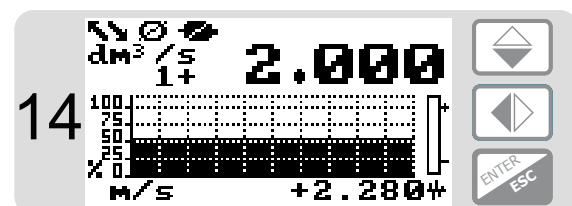
Press the ENTER button Confirm the new value



Press Esc



Press Esc



Main page

FUNCTIONS MENU

The main menu is selected from the Quick start menu by pressing the  key and entering the access code; enter the access code if required. Note: Some functions are displayed only with other functions active, or with optional modules.

SENSOR

MAIN MENU			
1	Sensor		
2	Units		
3	Scales		
4	Settings		
5	Help		
6	Exit		
7	Factory reset		
8	Calibration		
9	Zero point		
10	Flow rate		
11	Pressure		
12	Temperature		
13	Mass		
14	Specific gravity		
15	Linearization		
16	Flow sensor		
17	Pressure sensor		
18	Temperature sensor		
19	Mass sensor		
20	Specific gravity sensor		
21	Linearization sensor		
22	Flow sensor		
23	Pressure sensor		
24	Temperature sensor		
25	Mass sensor		
26	Specific gravity sensor		
27	Linearization sensor		
28	Flow sensor		
29	Pressure sensor		
30	Temperature sensor		
31	Mass sensor		
32	Specific gravity sensor		
33	Linearization sensor		
34	Flow sensor		
35	Pressure sensor		
36	Temperature sensor		
37	Mass sensor		
38	Specific gravity sensor		
39	Linearization sensor		
40	Flow sensor		
41	Pressure sensor		
42	Temperature sensor		
43	Mass sensor		
44	Specific gravity sensor		
45	Linearization sensor		
46	Flow sensor		
47	Pressure sensor		
48	Temperature sensor		
49	Mass sensor		
50	Specific gravity sensor		
51	Linearization sensor		
52	Flow sensor		
53	Pressure sensor		
54	Temperature sensor		
55	Mass sensor		
56	Specific gravity sensor		
57	Linearization sensor		
58	Flow sensor		
59	Pressure sensor		
60	Temperature sensor		
61	Mass sensor		
62	Specific gravity sensor		
63	Linearization sensor		
64	Flow sensor		
65	Pressure sensor		
66	Temperature sensor		
67	Mass sensor		
68	Specific gravity sensor		
69	Linearization sensor		
70	Flow sensor		
71	Pressure sensor		
72	Temperature sensor		
73	Mass sensor		
74	Specific gravity sensor		
75	Linearization sensor		
76	Flow sensor		
77	Pressure sensor		
78	Temperature sensor		
79	Mass sensor		
80	Specific gravity sensor		
81	Linearization sensor		
82	Flow sensor		
83	Pressure sensor		
84	Temperature sensor		
85	Mass sensor		
86	Specific gravity sensor		
87	Linearization sensor		
88	Flow sensor		
89	Pressure sensor		
90	Temperature sensor		
91	Mass sensor		
92	Specific gravity sensor		
93	Linearization sensor		
94	Flow sensor		
95	Pressure sensor		
96	Temperature sensor		
97	Mass sensor		
98	Specific gravity sensor		
99	Linearization sensor		
100	Flow sensor		
101	Pressure sensor		
102	Temperature sensor		
103	Mass sensor		
104	Specific gravity sensor		
105	Linearization sensor		
106	Flow sensor		
107	Pressure sensor		
108	Temperature sensor		
109	Mass sensor		
110	Specific gravity sensor		
111	Linearization sensor		
112	Flow sensor		
113	Pressure sensor		
114	Temperature sensor		
115	Mass sensor		
116	Specific gravity sensor		
117	Linearization sensor		
118	Flow sensor		
119	Pressure sensor		
120	Temperature sensor		
121	Mass sensor		
122	Specific gravity sensor		
123	Linearization sensor		
124	Flow sensor		
125	Pressure sensor		
126	Temperature sensor		
127	Mass sensor		
128	Specific gravity sensor		
129	Linearization sensor		
130	Flow sensor		
131	Pressure sensor		
132	Temperature sensor		
133	Mass sensor		
134	Specific gravity sensor		
135	Linearization sensor		
136	Flow sensor		
137	Pressure sensor		
138	Temperature sensor		
139	Mass sensor		
140	Specific gravity sensor		
141	Linearization sensor		
142	Flow sensor		
143	Pressure sensor		
144	Temperature sensor		
145	Mass sensor		
146	Specific gravity sensor		
147	Linearization sensor		
148	Flow sensor		
149	Pressure sensor		
150	Temperature sensor		
151	Mass sensor		
152	Specific gravity sensor		
153	Linearization sensor		
154	Flow sensor		
155	Pressure sensor		
156	Temperature sensor		
157	Mass sensor		
158	Specific gravity sensor		
159	Linearization sensor		
160	Flow sensor		
161	Pressure sensor		
162	Temperature sensor		
163	Mass sensor		
164	Specific gravity sensor		
165	Linearization sensor		
166	Flow sensor		
167	Pressure sensor		
168	Temperature sensor		
169	Mass sensor		
170	Specific gravity sensor		
171	Linearization sensor		
172	Flow sensor		
173	Pressure sensor		
174	Temperature sensor		
175	Mass sensor		
176	Specific gravity sensor		
177	Linearization sensor		
178	Flow sensor		
179	Pressure sensor		
180	Temperature sensor		
181	Mass sensor		
182	Specific gravity sensor		
183	Linearization sensor		
184	Flow sensor		
185	Pressure sensor		
186	Temperature sensor		
187	Mass sensor		
188	Specific gravity sensor		
189	Linearization sensor		
190	Flow sensor		
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192	Temperature sensor		
193	Mass sensor		
194	Specific gravity sensor		
195	Linearization sensor		
196	Flow sensor		
197	Pressure sensor		
198	Temperature sensor		
199	Mass sensor		
200	Specific gravity sensor		
201	Linearization sensor		
202	Flow sensor		
203	Pressure sensor		
204	Temperature sensor		
205	Mass sensor		
206	Specific gravity sensor		
207	Linearization sensor		
208	Flow sensor		
209	Pressure sensor		
210	Temperature sensor		
211	Mass sensor		
212	Specific gravity sensor		
213	Linearization sensor		
214	Flow sensor		
215	Pressure sensor		
216	Temperature sensor		
217	Mass sensor		
218	Specific gravity sensor		
219	Linearization sensor		
220	Flow sensor		
221	Pressure sensor		
222	Temperature sensor		
223	Mass sensor		
224	Specific gravity sensor		
225	Linearization sensor		
226	Flow sensor		
227	Pressure sensor		
228	Temperature sensor		
229	Mass sensor		
230	Specific gravity sensor		
231	Linearization sensor		
232	Flow sensor		
233	Pressure sensor		
234	Temperature sensor		
235	Mass sensor		
236	Specific gravity sensor		
237	Linearization sensor		
238	Flow sensor		
239	Pressure sensor		
240	Temperature sensor		
241	Mass sensor		
242	Specific gravity sensor		
243	Linearization sensor		
244	Flow sensor		
245	Pressure sensor		
246	Temperature sensor		
247	Mass sensor		
248	Specific gravity sensor		
249	Linearization sensor		
250	Flow sensor		
251	Pressure sensor		
252	Temperature sensor		
253	Mass sensor		
254	Specific gravity sensor		
255	Linearization sensor		
256	Flow sensor		
257	Pressure sensor		
258	Temperature sensor		
259	Mass sensor		
260	Specific gravity sensor		
261	Linearization sensor		
262	Flow sensor		
263	Pressure sensor		
264	Temperature sensor		
265	Mass sensor		
266	Specific gravity sensor		
267	Linearization sensor		
268	Flow sensor		
269	Pressure sensor		
270	Temperature sensor		
271	Mass sensor		
272	Specific gravity sensor		
273	Linearization sensor		
274	Flow sensor		
275	Pressure sensor		
276	Temperature sensor		
277	Mass sensor		
278	Specific gravity sensor		
279	Linearization sensor		
280	Flow sensor		
281	Pressure sensor		
282	Temperature sensor		
283	Mass sensor		
284	Specific gravity sensor		
285	Linearization sensor		
286	Flow sensor		
287	Pressure sensor		
288	Temperature sensor		
289	Mass sensor		
290	Specific gravity sensor		
291	Linearization sensor		
292	Flow sensor		
293	Pressure sensor		
294	Temperature sensor		
295	Mass sensor		
296	Specific gravity sensor		
297	Linearization sensor		
298	Flow sensor		
299	Pressure sensor		
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301	Mass sensor		
302	Specific gravity sensor		
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305	Pressure sensor		
306	Temperature sensor		
307	Mass sensor		
308	Specific gravity sensor		
309	Linearization sensor		
310	Flow sensor		
311	Pressure sensor		
312	Temperature sensor		
313	Mass sensor		
314	Specific gravity sensor		
315	Linearization sensor		
316	Flow sensor		
317	Pressure sensor		
318	Temperature sensor		
319	Mass sensor		
320	Specific gravity sensor		
321	Linearization sensor		
322	Flow sensor		
323	Pressure sensor		
324	Temperature sensor		
325	Mass sensor		
326	Specific gravity sensor		
327	Linearization sensor		
328	Flow sensor		
329	Pressure sensor		
330	Temperature sensor		
331	Mass sensor		
332	Specific gravity sensor		
333	Linearization sensor		
334	Flow sensor		
335	Pressure sensor		
336	Temperature sensor		
337	Mass sensor		
338	Specific gravity sensor		
339	Linearization sensor		
340	Flow sensor		
341	Pressure sensor		
342	Temperature sensor		
343	Mass sensor		
344	Specific gravity sensor		
345	Linearization sensor		
346	Flow sensor		
347	Pressure sensor		
348	Temperature sensor		
349	Mass sensor		
350	Specific gravity sensor		
351	Linearization sensor		
352	Flow sensor		
353	Pressure sensor		
354	Temperature sensor		
355	Mass sensor		
356	Specific gravity sensor		
357	Linearization sensor		
358	Flow sensor		
359	Pressure sensor		
360	Temperature sensor		
361	Mass sensor		
362	Specific gravity sensor		
363	Linearization sensor		
364	Flow sensor		
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366	Temperature sensor		
367	Mass sensor		
368	Specific gravity sensor		
369	Linearization sensor		
370	Flow sensor		
371	Pressure sensor		
372	Temperature sensor		
373	Mass sensor		
374	Specific gravity sensor		
375	Linearization sensor		
376	Flow sensor		
377	Pressure sensor		
378	Temperature sensor		
379	Mass sensor		
380	Specific gravity sensor		
381	Linearization sensor		
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383	Pressure sensor		
384	Temperature sensor		
385	Mass sensor		
386	Specific gravity sensor		
387	Linearization sensor		
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389	Pressure sensor		
390	Temperature sensor		
391	Mass sensor		
392	Specific gravity sensor		
393	Linearization sensor		
394	Flow sensor		
395	Pressure sensor		
396	Temperature sensor		
397	Mass sensor		
398	Specific gravity sensor		
399	Linearization sensor		
400	Flow sensor		
401	Pressure sensor		
402	Temperature sensor		
403	Mass sensor		
404	Specific gravity sensor		
405	Linearization sensor		
406	Flow sensor		
407	Pressure sensor		
408	Temperature sensor		
409	Mass sensor		
410	Specific gravity sensor		
411	Linearization sensor		
412	Flow sensor		
413	Pressure sensor		
414	Temperature sensor		
415	Mass sensor		
416	Specific gravity sensor		
417	Linearization sensor		